HR Attrition Data Analysis

Python-Based Exploratory Data Analysis

1. Project Overview

Employee attrition is an important business concern because employees leaving an organization can affect workforce stability and operational continuity. Understanding the patterns associated with employee attrition can help HR teams identify areas that require further investigation.

The objective of this project is to analyze employee attrition using Python, explore relationships between attrition and employee-related factors, identify high-risk employee groups, and provide data-driven recommendations to support employee retention efforts.

The analysis uses Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to inspect the dataset, calculate attrition rates, explore patterns, and visualize findings.

2. Dataset and Methodology

The dataset contains 1,470 employee records and 35 original columns, covering employee demographics, job-related information, compensation, satisfaction, work experience, and attrition status.

The analysis was performed using Python in a Jupyter Notebook. Pandas was used for data inspection, filtering, and aggregation; NumPy supported numerical operations; and Matplotlib and Seaborn were used to create visualizations.

The methodology included examining the dataset structure, checking missing values and duplicate records, calculating descriptive statistics, analyzing attrition across employee groups, exploring relationships between numerical variables, identifying potential income outliers, and developing visualizations to communicate the findings.

The analysis focused on identifying patterns associated with employee attrition and translating those findings into practical areas for HR to investigate.

3. Key Findings

The analysis revealed several patterns associated with employee attrition:

- **Overall attrition:** Out of 1,470 employees, 237 left the organization, resulting in an overall attrition rate of approximately 16.12%.
- **Overtime:** Employees working overtime had an attrition rate of 30.53%, compared with 10.44% among employees not working overtime.

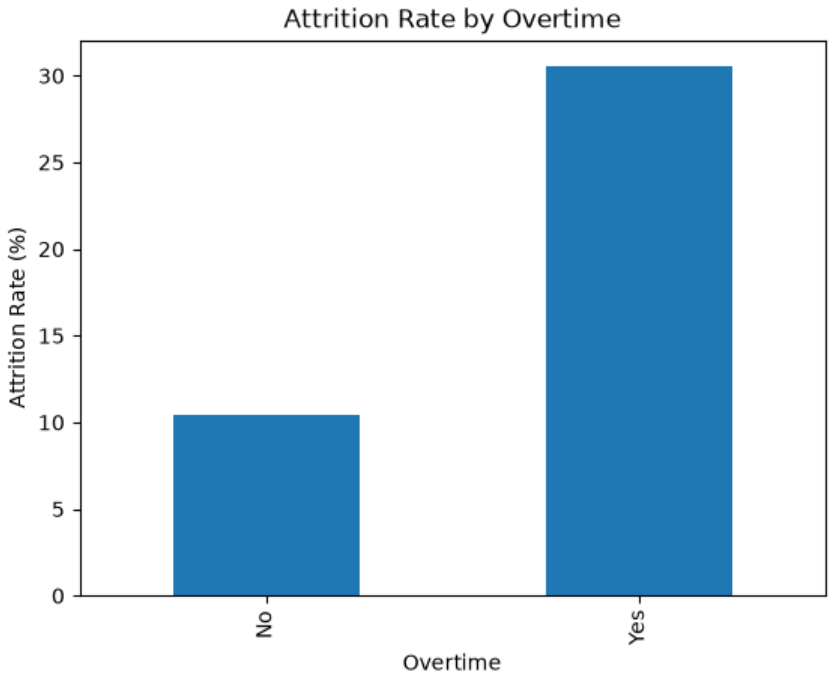


Figure 1: Employee Attrition Rate by Overtime Status

The chart shows that employees working overtime had a higher attrition rate (30.53%) than employees not working overtime (10.44%). This indicates an association between overtime and employee attrition.

- **Income:** The low-income group had an attrition rate of 29.27%, while the high-income group had a rate of 10.33%.

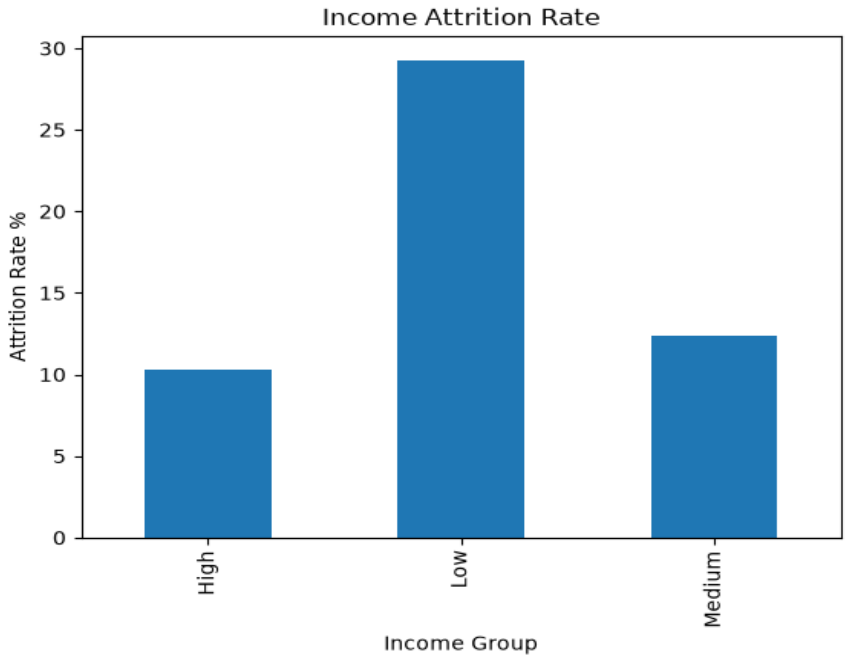


Figure 2: Employee Attrition Rate by Income Group

The low-income group had an attrition rate of 29.27%, compared with 12.41% for the medium-income group and 10.33% for the high-income group. This shows an association between income group and employee attrition.

- **Work-life balance:** Employees with the lowest work-life balance rating had an attrition rate of 31.25%.

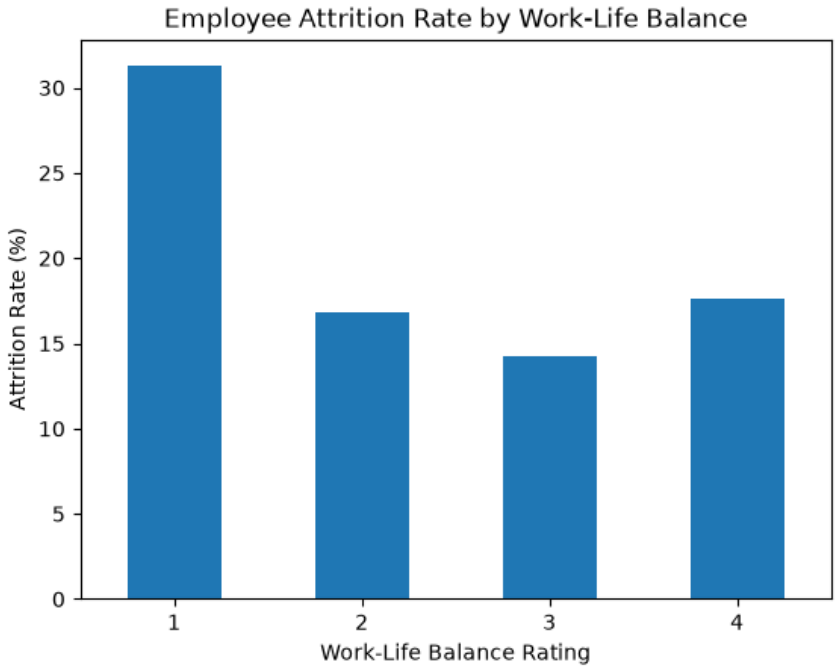


Figure 3: Employee Attrition Rate by Work-Life Balance

Employees with a Work-Life Balance rating of 1 had the highest attrition rate at 31.25%. The rate was 16.86% for rating 2, 14.22% for rating 3, and 17.65% for rating 4. This suggests an association between lower work-life balance ratings and higher attrition, although the pattern is not strictly linear.

- **Department:** Sales had the highest department-level attrition rate at 20.63%, followed by Human Resources at 19.05% and Research & Development at 13.84%.
- **Job level and overtime:** Overtime employees had higher attrition rates at all five job levels. The largest difference appeared at Job Level 1, where attrition was 52.56% among overtime employees compared with 15.76% among employees who did not work overtime.
- **High-risk subgroup:** Among 106 employees who belonged to the low-income group and worked overtime, 62 left the organization, representing an attrition rate of 58.49%.

These findings identify employee groups and workplace factors that warrant further investigation by HR. The observed patterns indicate associations and do not establish causation.

4. Recommendations and Conclusion

Recommendations

Based on the analysis, HR should prioritize investigating overtime practices, compensation, and work-life balance, as these factors were associated with higher employee attrition. Particular attention should be given to low-income employees who work overtime and employees at Job Level 1 who work overtime, as these groups showed comparatively high attrition rates.

HR should review workload distribution, compensation fairness, employee satisfaction, and work-life balance to understand the possible reasons behind these patterns. Retention initiatives should be developed based on the findings of further investigation, and attrition rates should be monitored over time to assess their effectiveness.

Conclusion

This project used Python-based exploratory data analysis to examine employee attrition patterns in a dataset of 1,470 employees. The analysis identified associations between attrition and factors such as overtime, income group, work-life balance, department, and job level.

The findings provide HR with a starting point for identifying employee groups that may need additional attention. However, the analysis is descriptive and does not establish that these factors cause employees to leave. Further investigation using additional evidence would help the organization make more informed employee-retention decisions.

Limitations

- The analysis is based on the available employee dataset and may not capture every factor influencing attrition.
- The findings show associations between employee characteristics and attrition; they do not establish cause-and-effect relationships.
- Some subgroup results are based on relatively small numbers of employees and should be interpreted cautiously.
- Further investigation and additional employee feedback may be needed to validate the findings and develop effective retention strategies.